Do Now · retrieval

Lesson 123 · Programming Paradigms: How We Express Solutions

Name: _____ Class: _____ Date: _____

Answer four retrieval questions before reviewing earlier learning.

Q1

What development approach is most explicitly risk-driven?

Q2

What is the effect of enqueue(X), enqueue(Y), dequeue() on an initially empty FIFO queue?

Q3

What is the time complexity of sequential search in the worst case?

Q4

After pushing A then B onto a stack, which is popped first?

New Learning · key ideas

The important knowledge from today's lesson. Read, annotate, and explain it.

Core idea 1

Imperative/procedural — Express a sequence of state-changing instructions and reusable procedures.

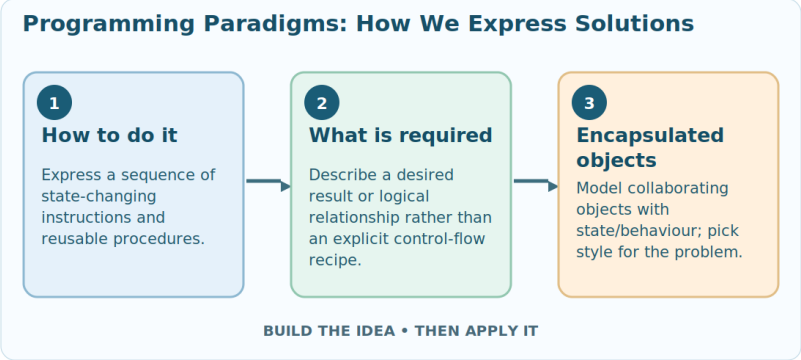
Core idea 2

Declarative — Describe a desired result or logical relationship rather than an explicit control-flow recipe.

Core idea 3

Object-oriented — Objects group state and behaviour. Inheritance lets a derived class reuse a base class; an override replaces a virtual method. Calling that method through a base-class reference runs the actual object's version: polymorphism .

Key visual · complete concept



Annotate the key visual

Explain which labels or features show the main process.

New Learning · applying the ideas

Vocabulary · use the correct technical terms

paradigm — Style or model for expressing computations.

imperative — Programming by explicit state-changing commands.

declarative — Programming by describing desired result/relations.

object-oriented — Programming using interacting objects with state and behaviour.

Check your understanding

Explain one key process in your own words using two of the terms above.

More animated examples are available in the original online lesson.

I Do · worked example

Problem

Compare a SQL query selecting all orders over £100 with a loop that scans each record.

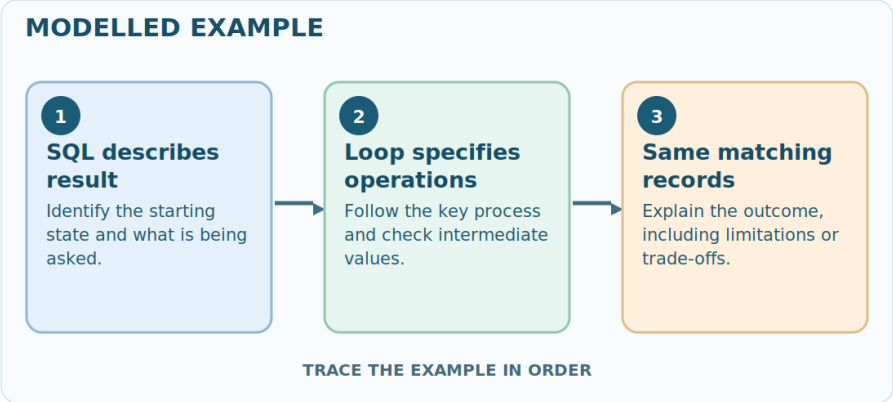
Follow the worked method

1. Model answer — SQL describes the required set of matching records (declarative); the loop specifies iteration, testing and append operations (imperative/procedural). Both can produce equivalent output through different expression styles. — Check: link each claim to the scenario and use precise OCR terminology.

2. Same signature, specialised behaviour — Show() is virtual in Alert . Both derived classes use override to supply their own implementation.

3. Trace the output — a.Show() prints Email then Popup . Although a is declared as Alert , the runtime selects the implementation belonging to the actual object. This is polymorphism, not two separate if tests.

Source model · final illustrated step



Why this method works

Identify the decision, rule or invariant that makes the method correct.

We Do · Try → Discuss → Improve

TRY independently in BLACK PEN; compare methods with a partner; improve your reasoning.

Guided problem · TRY

What must we decide?
Identify the inputs, relevant facts and exact output or judgement required.

DISCUSS

Compare how you reached the answer. Where could someone make an error?

IMPROVE

Add one precise correction or a stronger reason. Answers are not printed here.

You Do · independent practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 1

Define a programming paradigm and contrast imperative with declarative languages. [6]

Question 2

Explain how a SQL query differs conceptually from a loop-based filter. [6]

Check that you have used the technical terms and explained your decisions.

You Do · extended practice

Answer these selected original questions in BLACK PEN. Other exam tasks remain online.

Question 3

A base class Alert declares virtual Show(). Two derived classes EmailAlert and PopupAlert override Show(). An Alert[] contains one of each. Explain inheritance, overriding, polymorphism and one reason this design may be preferable to procedural type-checking. [8]

Check that you have used the technical terms and explained your decisions.

Exit Ticket · reflect and retrieve

Exit Ticket

Declarative programming always requires the programmer to specify every loop and update.

Use key vocabulary from memory

1. Explain paradigm in your own words.

2. Explain imperative in your own words.

3. Explain declarative in your own words.

Confidence and next step

How confident are you? Thumbs up / side / down

Which question would you improve after checking the classroom model?

Check answers after the lesson using the teacher-guided source models.